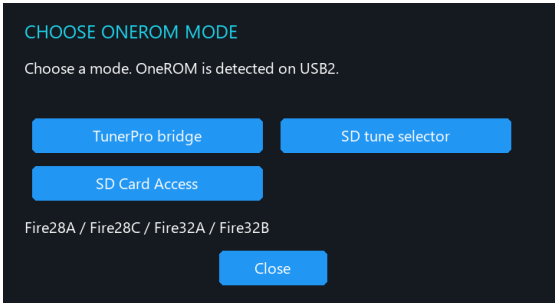
RAPID FIRE SOFTWARE

Rapid Fire Commander

Wired v0.13 handbook

Windows flashing, OneROM, TunerPro RT, SD tunes and wideband wiring.



Commander OneROM mode chooser

RapidFire-001 / rapid-fire-commander

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Print as a bifold booklet. In a PDF reader with booklet printing, choose **Booklet**, A4 paper, both sides and left binding. If the printer requires a flip setting, choose short-edge duplex. Fold the sheets together down the middle. The pages in this PDF are in reading order, so booklet printing handles the sheet order.

Overview and support

Rapid Fire Commander is a wired ESP32-P4 dashboard and OneROM tuning utility with read-only Holden ALDL logging through TunerPro RT.

Wired dashboard and OneROM tuning utility for the Guition JC4880P443C_I_W/Y 4.3-inch ESP32-P4 display. The Windows kit includes Commander v0.13, Espressif esptool and a guided flash launcher.

TunerPro + GT Pro mode provides separate COM ports for OneROM emulation and read-only Holden ALDL logging. Load the matching ADX in TunerPro RT. See the ADX logging setup.

The **SD Tune Selector** supports up to **64 `.BIN` files** from the root of a FAT32 microSD card. Only files that match the detected OneROM size can be selected.

Microsoft Edge download notice. SmartScreen may warn that a file isn't commonly downloaded when it has limited download history. If Edge or your antivirus shows a malware warning, please send the exact message and the file name or release version so I can investigate. If the GitHub page is unavailable in Edge, you can also view it in Firefox or Chrome. See Microsoft's SmartScreen guidance.

1. Download the Windows kit from the latest release.

2. Follow the Windows flashing guide for the board check, backup, bootloader sequence and first boot.

3. Use the menu and feature guide for OneROM, SD cards, TunerPro, wideband and display settings.

This repository distributes compiled Commander firmware and documentation. See the Commander firmware licence for copyright and permitted use. The Commander project source and tune files are private. The kit includes separate notices and source for the third-party flasher.

Use TunerPro RT through Commander's bridge for live calibration edits with compatible OneROM hardware. OneROM can stay installed in the vehicle's MemCal for SD tune swaps and full BIN uploads. Keep the engine off and verify the readback before starting it. See the TunerPro bridge and SD Tune Selector instructions.

Support

For help with this software, email Rapid Fire Software Support. Include the software version, what you were trying to do, and any error message you saw.

Print a folded copy

Print the whole PDF in Booklet mode on A4. Use both sides and left binding, then fold the sheets together. Keep this handbook with the Commander so the wiring and recovery steps are easy to find.

Flash on Windows

Check your board

This kit targets the **Guition JC4880P443C_I_W/Y**, a 4.3-inch, 480 × 800 touch display with an ESP32-P4, ST7701S LCD controller and 16 MB flash. The Commander build is for the early ESP32-P4 silicon range accepted by the image, v0.1 through v1.99. Use the exact board and verify its chip revision. Similar-looking Guition displays can have different panels, flash layouts or ports. Guition board specifications

Port	Use
USB2	Windows flashing in bootloader mode. In Commander, this port hosts the OneROM.
USB3	Windows TunerPro serial bridge or SD Card Access after boot.

Use a data-capable USB cable and Windows x64. Disconnect OneROM and any vehicle wiring for flashing. The merged image replaces the full 16 MB of P4 flash, including saved settings. Back up the original image first if you want a way back. This kit updates the P4 and leaves the separate wireless coprocessor firmware alone.

Enter the bootloader

1. Download the release ZIP and use **Extract All**. Keep the launcher, firmware and **tools** folder together.

2. Open **Device Manager > Ports (COM & LPT)** so you can see which port appears.

3. Unplug the USB and external power cables from the P4. Hold the board's **BOOT** button while reconnecting a Windows data cable to **USB2**. On the owner's tested unit this was the leftmost button. Use the board marking or vendor instructions on other revisions.

4. Release BOOT after Windows shows a new COM port. The LCD can stay blank in the bootloader. Write down the COM number. If several ports appear, compare the list with the board unplugged and re-enter bootloader mode.

Optional backup

Open PowerShell in the extracted folder. Replace `COM7` with your actual bootloader port. Pick a new filename and store the backup privately. Keep the P4 connected until the command completes.

```
.\tools\esptool.exe --chip esp32p4 --port COM7 --baud 115200 `
  --before no_reset --after no_reset `
  read_flash 0x0 0x1000000 p4-original-backup.bin
```

Flash and check the result

1. Double-click **Flash Commander.cmd**. Enter the P4's COM port. The launcher checks the firmware checksum and asks esptool to identify the chip, revision and flash size before offering to write.

2. Review the details. Type **FLASH** only for the intended board. The launcher writes the **merged** image at **0x0** and esptool checks the data it wrote. Keep the cable attached until the result is shown.

3. After success, release BOOT if you are still holding it. If the P4 stays in the bootloader, unplug and reconnect its power with BOOT released.

4. Look for the **Rapid Fire Commander** fade-in followed by the **Commander** dashboard. Verify touch response and check **Settings > Brightness**. Reconnect OneROM after the first boot check.

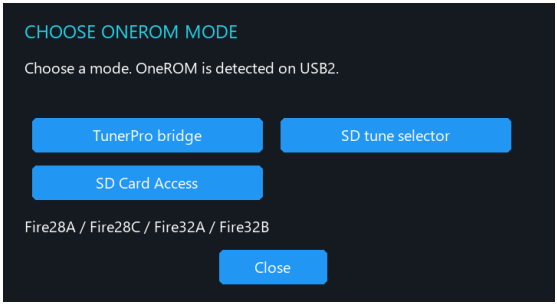
Troubleshooting

Symptom	What to check
No COM port	Use a known data cable directly to the computer. Check USB2 and repeat the BOOT sequence. Look in Device Manager for the new device.
Failed to connect or port busy	Close TunerPro, serial monitors and other flash tools. Confirm the COM number and repeat the BOOT sequence.
The launcher rejects the chip, revision or flash size	Compare the board label and chip identification with the compatibility section above.
Flash verification fails	Keep OneROM unplugged. Check cable and power, re-enter bootloader mode and retry the same kit.
Screen stays blank after success	Boot normally with BOOT released. If it stays blank, re-enter the bootloader, check the exact board model, and use the original backup for recovery if needed.
COM port disappears during normal boot	USB2 changes roles in Commander. Use USB3 and choose **TunerPro bridge** for a runtime COM port.

The kit uses the official Windows x64 esptool v4.9.1 release.

Commander menus and tuning

The illustrations are recreated from the firmware layout. Status values and example filenames illustrate the workflow.



OneROM mode chooser with TunerPro Bridge, SD Tune Selector and SD Card Access

Program and verify OneROM in the Rapid Fire OneROM app

1. Keep OneROM disconnected from the ECU. Connect it directly to your computer and open the Rapid Fire OneROM app.

2. The Rapid Fire OneROM app automatically detects and selects your OneROM board and sets the intended ROM size. Select the **NORMAL DT** firmware profile, then program the tune BIN you want to start with.

3. Read the programmed ROM back in the app and confirm that it matches the intended BIN byte for byte. If you plan to use the SD Tune Selector, put a copy of that verified BIN at the root of the FAT32 microSD card.

4. Once the readback matches, disconnect OneROM from the computer and plug it into Commander's **USB2** port. Open the Commander mode you want to use and confirm that it detects the expected board and ROM size.

OneROM modes

Mode	Use
TunerPro bridge	Laptop on USB3, OneROM on USB2. Select the P4's Ostrich COM port in TunerPro RT for emulation.
TunerPro + GT Pro	Separate Windows COM ports for OneROM emulation and read-only GT Pro ADX logging.
SD tune selector	Browse and load a matching BIN from the FAT32 microSD card onto OneROM with the engine off.
SD Card Access	Expose the P4's SD card as a removable drive through USB3. The tune selector and serial bridge pause while this mode owns the card.

TunerPro bridge

Connect the laptop to USB3 and the OneROM to USB2. Open **OneROM > TunerPro bridge**. Wait for the board and size to be detected, then find the new COM port in Windows and select it in TunerPro RT for Ostrich emulation. Start with a readback before attempting a write. Some OneROM profiles expose an optional second UART port. For GT Pro telemetry, choose **TunerPro + GT Pro**.

With a compatible OneROM setup and emulation enabled, TunerPro RT can send live calibration edits while the engine is running. TunerPro RT sends the editor update through Commander's bridge, and OneROM serves the updated data to the ECU. Perform live edits while the vehicle is stationary. See the TunerPro RT emulation guide.



TunerPro Bridge waiting for OneROM on USB2

TunerPro + GT Pro ADX logging

Commander v0.13 passes read-only ALDL requests and replies between TunerPro RT and the GT Pro. Load the ADX in TunerPro RT so it can decode the ECU replies.

1. Connect OneROM to **USB2** and the Windows laptop to **USB3**. On Commander, choose **OneROM > TunerPro + GT Pro**.

2. In **Settings > OBDX devices**, select the GT Pro and tap **Connect selected**. Wait for **GT Pro ALDL acquisition ready** on the OneROM screen.

3. In TunerPro RT, use the **OneROM emulation** COM port for Ostrich. Under **Preferences > Data Acq./Emulation**, choose **Standard Serial** and the separate **GT Pro ALDL acquisition** COM port. Windows assigns the port numbers.

4. Stop acquisition if it is running. Choose **Acquisition > Load Definition File**, load the ADX for your ECU and calibration, then start acquisition.

After a verified SD tune load, Commander can identify OSE $11P V104 or $12P V112. Use that recommendation to choose the matching ADX variant in TunerPro RT. For a factory or unknown profile, choose the definition for the installed ECU and calibration.

Software emulation passed 35 monitor polls across 27 Holden ADX definitions, including OSE $11P, OSE $12P, VS Enhanced and the stock VS V6 $51 profile. The owner also reported validating GT Pro Bluetooth and Windows COM port detection on the physical P4. See the Commander v0.13 release notes.

SD Tune Selector

The SD Tune Selector loads the selected BIN image in blocks. Even when two BINs share the same ECU program code, an ECU reading during the transfer can see a mix of old and new calibration data. For SD tune swaps and full BIN uploads, OneROM can stay installed in the vehicle's MemCal. Keep the engine off throughout the upload and verify the readback before starting it.

Fire model	Supported ROM size
Fire28A, Fire28C	16, 32 or 64 KiB
Fire32A, Fire32B	128 KiB, 27C010

Commander uses model and active-size metadata from OneROM. Use the matching OneROM v0.8.0 **NORMAL DT** firmware. Unsupported or unreadable profiles stop before a load.

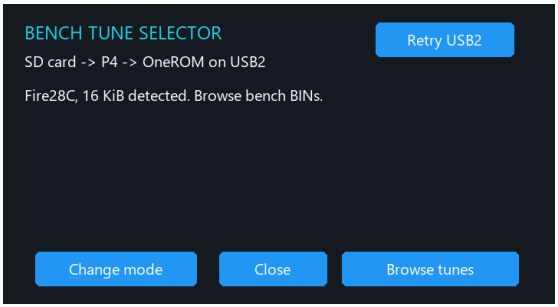
1. Copy the candidate BINs and a verified copy of the current BIN to the root of a FAT32 card. Use descriptive filenames.

2. Choose **SD tune selector** and check the detected board and ROM size.

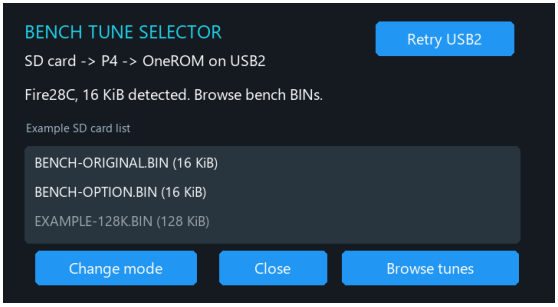
3. Tap **Browse tunes**. The list shows up to 64 BIN filenames and sizes. Files of a different size are disabled.

4. Select a compatible file and confirm the load. Commander checks the existing image against the card and validates the expected ROM layout before writing. For 32, 64 and 128 KiB images, identical halves fail this check.

5. Wait for **Verified byte for byte**. If a write is incomplete or verification fails, keep the engine off and restore and verify the original BIN before starting it using your OneROM recovery procedure.



Fire28C SD Tune Selector ready to browse



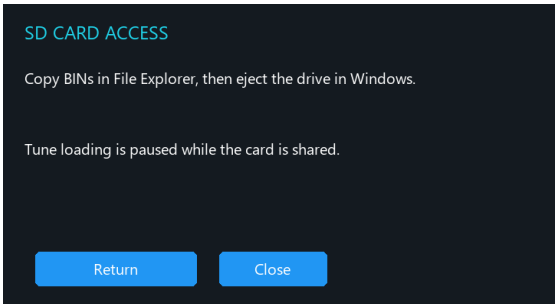
Example SD BIN list with a disabled size mismatch

Tap **Retry USB2** if the selector loses the device. The P4 can restart its USB2 host and the OneROM session while the cable stays in place. A retry does not

replay a write. For **Change mode**, finish transfers, close TunerPro, turn the engine off and confirm the restart. A restart can discard unsaved tune changes.

SD Card Access

Choose **SD Card Access** and connect Windows to USB3. Copy files to the card in File Explorer. Eject the drive in Windows and wait for the P4 to show that the card was released. Tap **Return** and select the next mode after the restart. You can also copy files with a separate card reader.



SD Card Access with Windows eject instructions

Dashboard, Engine and Settings

• **Dashboard** and **Engine** show speed, RPM, intake air, coolant, spark advance and battery fields. Unavailable readings display dashes.

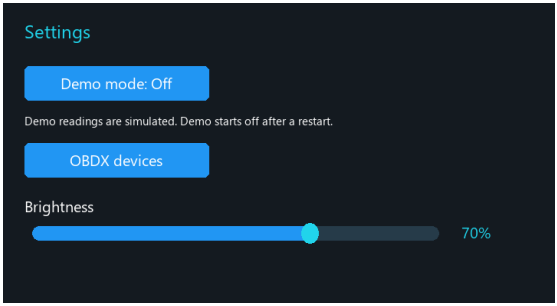
• **Settings > Brightness** adjusts the backlight from 10% to 100% and saves on release.

• **Settings > Demo mode** supplies simulated gauge values, marked by a **DEMO** badge. It starts off after a restart.

• **Settings > OBDX devices** opens Bluetooth discovery, connection controls and diagnostics. Use **TunerPro + GT Pro** to log ECU data in TunerPro RT.

• **Wideband > Source** chooses Innovate LM-2 or AEM 30-0300. See wideband setup for wiring and signal levels. Use one source at a time.

Settings and startup screens



Settings with Demo mode and brightness



Rapid Fire Commander startup logo fade

Wideband wiring

Commander receives a digital wideband signal on **P4 UART2 RX, GPIO32**. Use one supported source at a time. The screen shows AFR and lambda when valid data arrives.

Find the wideband input on the display board

On the Guition **JC4880P443C_I_W** schematic, **JP1** is the 2×13 header labeled **Expand IO**. Look for **GPIO32** at JP1 pin 19 and **GND** at pin 5. The schematic shows pin 20 beside pin 19 as **C6_U0RXD**. That pin belongs to the ESP32-C6. The separate RS-485 connector uses different signals. Match the board model and pin-1 orientation to the drawing before wiring. Guition Expand IO schematic

Required serial interface

Use a **true RS-232 to 3.3 V UART receiver**, for example a MAX3232 breakout designed for 3.3 V operation and powered at 3.3 V. Check the breakout pinout for its RS-232 receive input and its 3.3 V logic receive output. Keep raw RS-232 voltages, 5 V logic, vehicle power, CAN and analog outputs away from the P4 GPIO. A USB-to-TTL cable alone does not convert RS-232 signal levels. MAX3232 reference

Signal	Wiring path
RS-232 data	Controller TX > R1IN > R1OUT > JP1 pin 19 (GPIO32)
Ground	Controller signal ground > receiver ground > JP1 pin 5 (GND)

From	To
Wideband RS-232 TX	Receiver RS-232 input, often labeled R1IN
Receiver 3.3 V logic output, often labeled R1OUT	Guition JP1 pin 19, P4 GPIO32
Receiver logic ground	Guition JP1 pin 5, GND

From	To
Wideband serial signal ground	Receiver RS-232 signal ground, according to the controller and receiver pinouts

The P4 connection is receive-only. Leave adapter transmit wiring to the wideband unused. Power the module from a suitable regulated 3.3 V source as specified by the module. Check the JP1 orientation and actual board revision with power off. Guition board specifications

Choose the source

Controller	Signal	Commander setting
Innovate LM-2	MTS Serial OUT. Use the matching MTS serial-out lead for the LM-2 connector and verify its TX and ground pinout.	**Wideband > Source: LM-2**. The firmware listens at 19,200 baud, 8N1 and uses the first channel.
AEM X-Series 30-0300	Blue RS-232 Serial OUT wire. Set the gauge to AFR display mode.	**Wideband > Source: AEM**. The firmware listens at 9,600 baud, 8N1 and derives lambda using a gasoline reference of 14.65.

Compare the P4 reading to the controller display on the bench. Dashes indicate unavailable input or invalid readings. The dashboard returns to LM-2 after a reboot. This feature displays readings on the P4. TunerPro wideband passthrough remains pending, and both sources still need physical verification on this build. Innovate LM-2 manual · AEM 30-0300 manual

Firmware licence

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This licence covers the original Commander firmware. Bundled third-party software, including Espressif esptool, and third-party firmware components are governed by their respective licences and notices.

Commander v0.11 was flashed and run on the owner's Guition 4.3-inch ESP32-P4. Follow the Windows flashing guide and back up the existing firmware. For OneROM writes, follow the bench procedure with the ECU disconnected.

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